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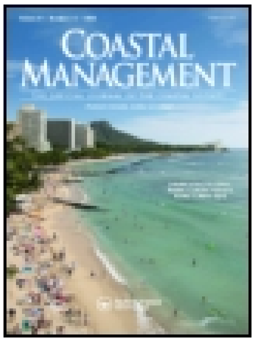

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Conceptualizing Social Outcomes of Large Marine Protected Areas

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ABSTRACT
There has been an assumption that because many large marine protected areas (LMPAs) are designated in areas with relatively few direct uses, they therefore have few stakeholders and negligible social outcomes. This article challenges this assumption with diverse examples of social outcomes that are distinctive in LMPAs. We define social outcomes as inclusive of both *social change processes* and *social impacts*, where "social" includes all perceptual or material human dimensions. We draw on five in-depth case studies to report social outcomes resulting from proposed or designated LMPAs in Bermuda, Rapa Nui (Easter Island), Kiribati, Palau, and the Commonwealth of the Northern Mariana Islands & Guam. We conclude: (1) social outcomes arise even in remote LMPAs; (2) LMPA efforts generate social outcomes at all stages of development; (3) LMPAs have the potential to produce outcomes at a higher level of social organization, which can change the scope and type of affected populations and, in some cases, the nature and stakes of the outcomes themselves; (4) the potential for LMPAs to impart distinctive social outcomes results from their unique geographies and/or intersection with high-level politics and policy processes; and (5) social outcomes of LMPAs may emerge in the form of social change processes and/or social impacts.

KEYWORDS
human dimensions; large marine protected areas; marine conservation; social outcomes

Introduction

Large marine protected areas (LMPAs) greater than 100,000 km^2 are a significant contemporary trend in marine conservation, now accounting for more than half of the global ocean area under protection (Juffe-Bignoli et al. 2014). As social-ecological systems, LMPAs are

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‡All authors contributed significantly. RG coordinated and led paper conceptualization and writing; she also contributed to data collection and analysis. LF contributed substantially to data analysis and writing. LA, EA, LMC, NJG, LM, and SBZ made equal contributions in terms of data collection, analysis, and/or writing. KW contributed to the analysis of the MTMNM case.

different from smaller, coastal MPAs (hereafter: conventional MPAs) in form, function, and/or conceptualization (Gruby et al. 2016; Wagner 2013). They often encompass remote and open ocean areas and islands, including pelagic and deep benthic ecosystems not typically incorporated in conventional MPAs (Toonen et al. 2013; Wilhelm et al. 2014). Due in part to the size and remoteness of LMPAs, they may also impart unique human uses, values, politics, governance arrangements, and outcomes (Gruby et al. 2016).

The human dimensions of LMPAs may also be presumed different from conventional MPAs, or even presumed irrelevant (Gruby et al. 2016). In recent years, for example, a discourse of LMPAs as protecting "the final marine wilderness areas" (Graham and McClanahan 2013, 397) has informed assumptions that LMPAs will be "politically and administratively easier to establish" (Balmford et al. 2004, 9696) than conventional MPAs because they are considered to have "few vested interests and stakeholders" and therefore negligible social impacts (Singleton and Roberts 2014, 9). Social scientists have begun to challenge these assumptions, arguing that LMPAs, similar to conventional MPAs, are likely to have significant but potentially unique social outcomes (e.g., Gruby et al. 2016). But what are those outcomes, and how are they similar to and different from those observed in conventional MPAs? Empirical research on the range, magnitude, and distribution of social outcomes of LMPAs has been identified as a priority (Christie et al. 2017; Gruby et al. 2016).

This paper draws on five in-depth case studies of proposed and designated LMPAs in Bermuda; Rapa Nui (also known as Easter Island, Chile); the Republic of Palau (Palau), the U.S. Commonwealth of the Northern Mariana Islands (the CNMI) and U.S. Territory of Guam, and the Republic of Kiribati (Kiribati) to highlight examples of social outcomes that are distinctive in LMPAs. In focusing on distinctive outcomes, we are not suggesting that LMPA do not also exhibit social outcomes common in conventional MPAs. Emerging work is examining the extent to which social outcomes common in small-scale MPAs are also arising in LMPAs (e.g., Ban et al 2017). Our contributions here are, first, to establish the need for researchers and practitioners to look for social outcomes that are less familiar and, second, to offer a conceptual framework that may be helpful in doing so.

Conceptualizing social outcomes

Despite their importance, the social outcomes of protected areas "remain significantly underresearched" (Jones, McGinlay, and Dimitrakopoulos 2017, 1). This includes MPAs, where research is needed on the magnitude, distribution, and temporal variance of social impacts (Fox et al. 2012, 3). West et al. (2006, 265) caution against simplistic assessments that condense "rich and nuanced social interactions […] to a few easily conveyable and representable issues or topics." Here we engage the literature on social impact assessment (SIA) for a conceptual framework that enables us to characterize and parse a broad and complex suite of social outcomes in LMPAs.

SIA is an applied field with origins in the U.S. National Environmental Policy Act (Esteves, Franks, and Vanclay 2012). SIAs are typically conducted as part of required environmental impact statements to predict social impacts and inform decision-making (Esteves, Franks, and Vanclay 2012; Tilt, Braun, and He 2009). However, SIA can also be used as a scholarly "research tool" (Tilt, Braun, and He 2009) to understand impacts resulting from interventions (Vanclay 2012, 150). SIA has been underutilized in the fields of marine conservation and management (Vanclay 2012; Voyer, Gladstone, and Goodall 2012).

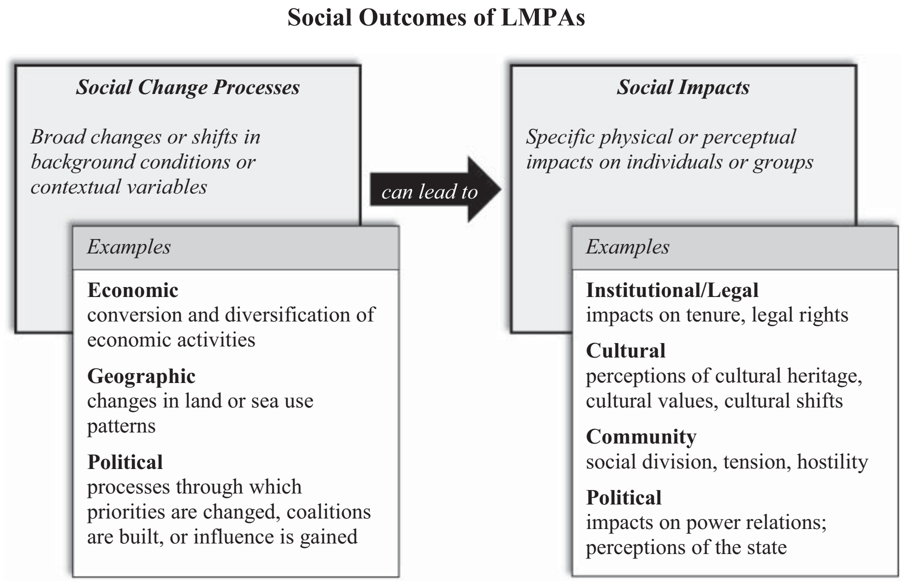

Figure 1. Conceptual framework for parsing the social outcomes of LMPAs, highlighting examples illustrated by our case studies (informed by Vanclay 2002).

We did not conduct SIAs in the traditional sense. Rather, we engage the SIA literature to outline a conceptual framework that defines social outcomes as inclusive of both *social change processes* and *social impacts* (see Figure 1). We emphasize that 'social' is used in the SIA literature and herein broadly, spanning every conceivable social dimension (e.g., cultural, demographic, economic, health, institutional, political) (Vanclay 2002). Social change processes are "intervening variables that might lead to social impacts under certain conditions" (Vanclay 2002, 207). Displacement, for example, is a geographical social change process that can be initiated by a no-take MPA in a heavily fished space. Social impacts follow from social change processes. These are "the impacts actually experienced by humans (at individual and higher aggregation levels) in either a corporeal (physical) or cognitive (perceptual) sense" (Vanclay 2002, 191). Impacts may be direct or indirect, intended or unintended, positive or negative, and they may be experienced at any level of social organization (Vanclay 2002). Continuing the earlier example, displacement caused by a no-take MPA may result in social impacts like declines in human health or economic wellbeing. For Vanclay (2002) social impacts are the ultimate outcome of interest, with social change processes important mainly as "intervening variables" that affect whether and how social impacts emerge. Here, we argue that social change processes and social impacts are both forms of social outcomes that are distinct and worthy of attention in their own right. In presenting results, we replace the generic "social" with a case-specific descriptor to reference particular types of social impacts and social changes processes highlighted by our cases (e.g., political impacts or political change processes). We use 'social outcomes' to refer generally to both social impacts and social change processes.

Research on conventional MPAs has demonstrated a wide range of social outcomes (see Bennett and Dearden 2014 for brief review). This research has attended to both social impacts and social change processes through the language of: welfare (Mascia, Claus, and Naidoo 2010), well-being (Christie 2004), social impacts (Fox et al. 2012), processes-oriented factors (Christie 2004), and benefits and costs (Charles and Wilson 2009). Most of this literature has focused on the social outcomes of MPAs within coastal communities, particularly for fishers but also other resource users like recreational users and recreational service providers (e.g., charter boats, dive companies) (Bennett and Dearden 2014; Charles and Wilson 2009; Hattam et al. 2014; Mascia, Claus, and Naidoo 2010; Rees et al. 2013; Smith et al. 2010). Mascia, Claus, and Naidoo (2010), for example, assess the social impacts of MPAs on local food security, resource rights, employment, community organization, and income. Others have highlighted nonmaterial social impacts such as social capital (Nenadovic and Epstein 2016), environmental awareness (Walmsley and White 2003), or unmet expectations (Fox et al. 2012). The literature also addresses social change processes triggered by MPAs, including those related to stakeholder participation and conflict-resolution (Christie 2004), decision-making authority and resource use rights (Gruby and Basurto 2013), and social conflict (Jones 2009).

Research on conventional MPAs underscores the importance of taking social outcomes into account when designing and implementing MPAs for both instrumental and ethical reasons. Demonstrating tangible social benefits can be the “deciding factor” in public support for MPAs (Charles and Wilson 2009, 11), which may ultimately determine the fate of fisheries and biodiversity conservation outcomes (Walmsley and White 2003). Understanding and mitigating negative social impacts is equally important for improving social acceptance of MPAs and reducing conflict, which, again, is instrumental to ecological success (Ranger et al 2016). In addition to effectiveness, there is a strong ethical argument for ensuring that MPA outcomes are equitable (Spalding et al. 2016) and socially just (Jones 2009).

It is equally important, for both instrumental and ethical reasons, to understand the social outcomes of LMPAs. Emerging research suggests that impacts observed in conventional MPAs, relating to livelihood benefits, opportunity costs for fishers, social injustice, and social conflict may present with LMPAs as well (e.g., Ban et al. 2017; De Santo 2013; Harris 2014; Richmond and Kotowicz 2015). However, there are still competing viewpoints (e.g., Jones and De Santo 2016; Singleton and Roberts 2014), many questions, and few empirical studies examining the unique social outcomes of LMPAs. Do LMPAs direct attention and resources away from other conservation approaches (Dulvy 2013)? Do they enable states or indigenous peoples to strengthen their sovereignty over marine spaces in ways that smaller MPAs would not (Leenhardt et al. 2013)? Can they provide opportunities to protect cultural seascapes (Wilhelm et al. 2014)? Are they reconfiguring national economies (Gruby et al. 2016)? Although we do not answer all of these questions, we do address the broader question underlying them: what distinct social outcomes are arising in LMPAs, and why?

Methods

Our case study sites (Figure 2) include: a stalled attempt to designate an LMPA in Bermuda (Bermuda Blue Halo Initiative); an LMPA proposal currently under consideration in Rapa Nui[1]; a recently declared LMPA in Palau and two longer established LMPAs in Kiribati and in the CNMI and Guam. The term ‘LMPA’ is not used locally in most of our study sites. We use ‘LMPA’ here as an organizing concept for conservation in marine areas larger than 100,000 km^2 (Spalding et al. 2014), which includes all of our cases. We selected these cases in part to explore outcomes that have arisen in

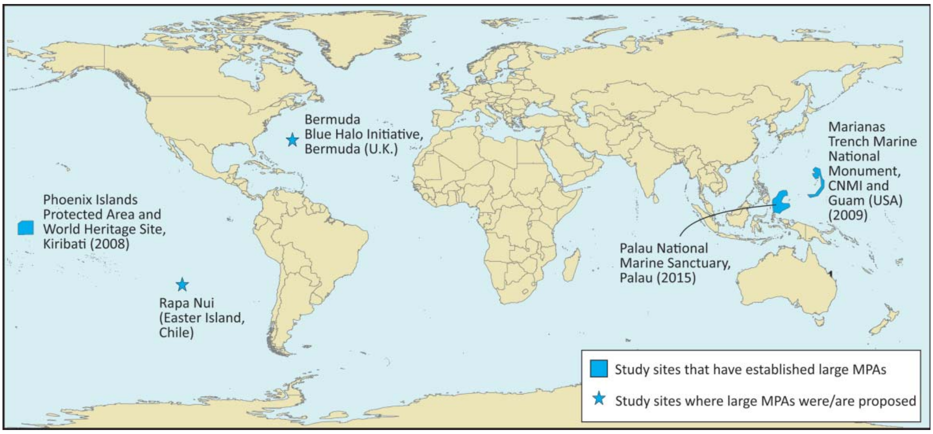

Figure 2. Case study locations.

LMPAs at varying stages. While our cases vary in stage of development, all are associated with small islands.

We conducted a combined 20 months of fieldwork, collecting data in key centers of decision-making (both local and overseas) for each of the case studies (see Table 1). For all cases, we used semi-structured interviews, participant observation, and document analysis to collect and triangulate data. Interviewees included representatives from governments, local and global NGOs, traditional leaders, ocean user and industry groups, researchers and contractors, and informed members of the public (see Table 2). We rely more or less on different data sources depending on the case. For example, our fieldwork in Rapa Nui coincided with ongoing deliberation processes; as a result, this case study is informed more by participant observation than interviews relative to others. We used a combination of purposeful and snowball sampling to identify interviewees who were either closely connected to the LMPA process and/or part of a key stakeholder group.

We collected data on both perceived and material social outcomes, although our focus was on the former. Our reliance on perceptions is consistent with the predominant method social scientists use to evaluate social impacts of protected areas (de Lange, Woodhouse, and Milner-Gulland 2017) and recognition of the importance of perceptions as evidence for

Table 1. Fieldwork summary.

Case study	Stage of development	Data collection sites	Time in the field	Number of interviewees
Rapa Nui (Easter Island, Chile)	Under consideration beginning in 2015	Rapa Nui and Chile	4 months between 2015–2016	30
Bermuda Blue Halo Initiative	Not designated after discussions between 2010–2015	Bermuda, U.K., Washington, D.C.	5.5 months, 2014–2016	104
Palau National Marine Sanctuary	Designated* in 2015	Palau	4 months between 2015 and 2017	84
Phoenix Islands Protected Area	Designated in 2008	Kiribati	3 months in 2016	47
Marianas Trench Marine National Monument	Designated in 2009	Commonwealth of the Northern Mariana Islands, Guam, Hawai'i	3.5 months between 2015–2016	50

*By designated we mean formally established through legal means.

Table 2. Total number of project interviewees and their primary roles.

Primary role	Number of interviewees
NGO—Global	30
NGO—Site based	46
Industry—Industrial and small-scale fishing, Tourism, Other	53
Traditional leaders	3
Government—Territorial mainland (U.S., U.K., Chile)	17
Government—Foreign	6
Government—Site based	110
Researchers and contractors	39
Informed citizens and others	13
Total	317

Note: Table 2 includes two interviewees not captured in Table 1 because they work at a global scale and are not associated with one particular case study site.

understanding conservation more broadly (Bennett 2016). The SIA field also emphasizes the importance of understanding perceived impacts:

> Perceived impacts are real social impacts. Although some scientists and engineers might discount people's perceptions because they are not real and may be based on incorrect information, the important social understanding is that, right or wrong, the way individuals perceive things affects their feelings. […] People act on their perceptions and people feel as a result of their perceptions. (Vanclay 2012, 152)

In order to understand social outcomes, we systematically collected data across the cases through the use of consistent interview guides with probes about cultural, economic, political, environmental, and institutional processes and impacts, which have been identified as being important in LMPAs (see Gruby et al. 2016). However, through the use of semistructured interviews we also remained open to the discovery of different types of social outcomes. This approach is consistent with the SIA literature, which cautions against the use of "checklist" approaches that blind analysts to unforeseen outcomes (Vanclay 2012).

We transcribed interviews and uploaded transcripts, policy documents, and participant observation notes to QSR NVivo for analysis. We coded data using a shared coding framework attentive to social outcomes that were experienced, promised, and expected; the processes through which outcomes emerged; and the way interviewees interpret those processes and outcomes. We enhanced intercoder reliability—the degree of agreement among multiple coders—through several rounds of test coding through which the authors[2] independently coded the same interview transcript and then collectively discussed areas of disagreement. Through this iterative process, we refined code names and descriptions, enhanced our collective understanding thereof, and developed shared coding rules. Unless specifically noted otherwise, all results presented in this paper are derived from analysis of primary data.

Though we found a broad range of social outcomes associated with each case, the selection of outcomes we examine here was informed by our objective to draw attention to social outcomes in LMPAs that are not common in conventional MPAs. The remainder of the paper will focus on five key social outcomes—one for each case—that meet this criteria. It is beyond the scope of this paper to report the full range of social outcomes arising in our case studies. These will be described in future work.

Given the relatively narrow focus of this article on distinctive social outcomes, we also emphasize that we do not necessarily focus on outcomes that we and/or informants may consider the most significant for each case. For example, one of the most salient social impacts of LMPA efforts in Bermuda and the CNMI was conflict. Interviewees in Bermuda described new divides in the marine conservation community, erosion of trust in the government, and increased distrust in foreign or "outsider" organizations as a result of the Blue Halo process. In the CNMI, many interviewees identified "unfulfilled promises" as the main outcome of the MTMNM, reflecting longstanding tensions about the LMPA within the CNMI. Another salient outcome in several of our case studies (Palau, the CNMI, and Kiribati) was the expectation of economic impacts linked to tourism. All of these outcomes are common within conventional MPAs, and protected areas more broadly (West, Igoe and Brockington 2006). Thus, here we are deliberate about focusing on examples from our cases that can increase awareness of other types of social outcomes that may be associated with LMPAs.

Results

Bermuda Blue Halo Initiative (Bermuda)

Bermuda is a UK Overseas Territory in the Atlantic Ocean. An LMPA has not been established in Bermuda and, as of this writing, appears unlikely to be established in the near future, despite a 6-year political effort to do so. The Bermuda LMPA effort emerged from, impacted, and interacted with an international initiative, later named the Sargasso Sea Alliance (SSA), that Bermuda led to promote conservation in the Sargasso Sea. The negotiations over a potential LMPA in Bermuda's Exclusive Economic Zone (EEZ) increased stakeholder concerns and conflict over its (potential) protected status, which is an important community impact in Bermuda. Here, however, we focus on a 'spillover' outcome related to the conflict over the LMPA in Bermuda: an institutional impact affecting an international conservation process for Sargasso Sea conservation.

A concern for global oceans conservation, the Sargasso Sea is a large gyre encompassing much of the north-central Atlantic Ocean. The boundaries of the Sargasso Sea encompass a broad swath of international waters as well as Bermuda's EEZ, as delineated by the SSA and further specified through a workshop coordinated by the Convention on Biological Diversity (Freestone 2014). Between 2012 and 2014, the terms of the international collaboration were negotiated as a non-binding political statement called the Hamilton Declaration on Collaboration for the Conservation of the Sargasso Sea. Under the proposed Hamilton Declaration, Bermuda would retain full authority over the governance of its EEZ, but as a signatory would agree to collaboratively pursue conservation for the Sargasso Sea through existing regional and international organizations (Hamilton Declaration 2014, 3). For example, signatories might support stronger shipping regulations through the International Maritime Organization or propose fisheries policy that would protect vulnerable species or ecosystems through the UN Food and Agriculture Organization (Hamilton Declaration 2014, 4). The declaration itself did not specify the conservation measures to be taken.

In 2010, during the initial stages of negotiations on Sargasso Sea conservation, a subset of negotiators facilitated a collaborative agreement between Pew Charitable Trusts (Pew) and the Bermudian government to advance the designation of an LMPA in Bermuda's EEZ. The LMPA was initially proposed as a way to demonstrate Bermuda's commitment to large-scale marine conservation as the leader of the Sargasso Sea effort. The later proposed LMPA consisted of a no-take ring, or "Blue Halo," extending from 50 miles off Bermuda's coast to the

200-mile EEZ limit around Bermuda. In preparation for a public consultation about the EEZ, Pew staff promoted the Blue Halo through media and community campaigns. As the visibility of the Blue Halo initiative grew, some stakeholder groups, such as fishermen and deep sea mining investors, began to voice concerns about placing restrictions within a large portion of Bermuda's EEZ. These included economic and governance concerns, as well as concerns about Bermuda's autonomy over its own waters given the campaign led by a foreign NGO, among others. From August to October of 2013, Bermuda's Sustainable Development Department conducted a public consultation process to broadly explore how to govern Bermuda's EEZ. The public divisiveness of the Blue Halo initiative came to the fore during the consultation and the months that followed.

As the signing of the Hamilton Declaration approached in early 2014, the Bermudian government had not yet made a decision on how to govern its EEZ, and the potential Blue Halo LMPA remained controversial. This controversy influenced debates about the Hamilton Declaration and the inclusion of Bermuda's EEZ in the Sargasso Sea Geographical Area of Collaboration. Some interviewees expressed concern that including Bermuda's EEZ implied some loss of Bermudian authority over its own waters given the participation and influence of foreign NGOs, particularly the Waitt Foundation, in the Sargasso Sea conservation effort.

Just days before the Hamilton Declaration was signed in 2014, a small group of Bermudians met with Premier Craig Cannonier to voice their concerns about including Bermuda's EEZ in the Sargasso Sea Geographical Area of Collaboration specified in the Hamilton Declaration. The Premier ultimately announced that, while Bermuda would sign the Hamilton declaration to support conservation in the Sargasso Sea outside of Bermuda's EEZ, Bermuda would pull its EEZ out of the Area of Collaboration, reducing the overall conservation area by about 437,000 km^2 (or 10.5% of the total).[3] Premier Michael Dunkley later suspended formal negotiations about the Blue Halo LMPA until 2017 at the earliest.

In summary, while an LMPA has not been designated in Bermuda, negotiations over the Blue Halo resulted in community impacts of increased concerns and tension over the role of Bermuda's EEZ in oceans conservation. This controversy contributed to Bermuda's withdrawal of its EEZ from the Hamilton Declaration, which we understand as an institutional impact that greatly reduced the area included within the Sargasso Sea conservation effort. The case of a proposed LMPA in Bermuda thus demonstrates the various scales, from local to international, at which an LMPA proposal can have unintended outcomes.

Rapa Nui (Easter Island, Chile)

Rapa Nui (Easter Island) is a special territory of Chile. The local people, the Rapanui, identify strongly with their Polynesian origins and are legally recognized by Chile as indigenous people. In its history of colonization, Rapa Nui was annexed as a part of Chile in 1888 and Chile recognized the rights of the Rapanui as Chilean citizens in 1966. Chile is signatory to both the Indigenous and Tribal Peoples Convention, 1989, and the UN Declaration on the Rights of Indigenous Peoples; the former requires the state to consult indigenous peoples on any actions taken in indigenous territory. The relationship between Chile and the Rapanui is central to marine conservation efforts in Rapa Nui, where an emergent dialogue between indigenous rights and marine conservation has shifted the focus of large-scale marine conservation to center on questions of territorial rights to the sea. As one of the cases in our study where an LMPA has not yet been established, Rapa Nui provides insights into outcomes related to the process of discussing a possible LMPA.

The local community in Rapa Nui has observed declines in marine resources over the past three decades (Gaymer et al. 2013). Community members, local NGOs, international NGOs, universities, and the Chilean government have discussed multiple options for marine conservation for Rapa Nui, beginning in 2010. Although various visions for marine conservation have been discussed, in October of 2015 at the Our Ocean conference hosted by Chile, Chilean President Michelle Bachelet announced the intention to establish an LMPA in Rapa Nui, pending the approval of the Rapanui people. The government began planning a marine conservation consultation process, but this was postponed when a different, more urgent indigenous consultation unfolded around the Special Statute for Rapa Nui. The Special Statute has been proposed and discussed since at least 2005. The current consultation attempts a renegotiation, between Chile and the Rapanui, of: the governance of the island, the administration of territorial land, and the administration of the territorial sea. In contrast, previous attempts at consultations for the Special Statute of Rapa Nui only included terrestrial territory; it did not include questions of administration of the sea. Since the most recent Special Statue negotiations included the topic of the sea, conversations concerning the LMPA became entangled in discussions about indigenous rights and Rapa Nui's relationship with Chile.

Some community members we interviewed suggested that the conversations concerning a possible LMPA in Rapa Nui put pressure on the government to conduct the Special Statute consultation, because the rights and control over marine space in Rapa Nui needed to be specified first to provide clarity for the potential marine conservation area. Others have hypothesized that because the Special Statute and marine conservation consultations were being suggested concurrently, the Chilean government combined the indigenous consultations needed for both topics in order to be efficient. Either way, the intersection of the LMPA consultation and the Special Statute is indicative of a political change process that has shifted the terms and conditions of the marine conservation discussion in Rapa Nui.

Although initial proposals for marine conservation in Rapa Nui centered around the possibility of creating an LMPA, the Special Statue centered on the establishment of indigenous rights to the entire 200-mile EEZ as a precursor to any marine conservation. While the Special Statute consultation is currently suspended, its effects linger as the discussion concerning marine conservation in Rapa Nui moves forward. The emergent dialogue between marine conservation and the Special Statue shifted the focus of many marine conservation groups in Rapa Nui toward rights to and control over the sea. The Special Statute discussion itself was also transformed by the interaction with the LMPA process; prior discussions of territorial administration in the Special Statute did not include discussions of the administration of the sea. In this way, the encounter between the potential LMPA and the Special Statute expanded claims of Rapanui territory beyond the land to include the entire 200-mile marine EEZ surrounding Rapa Nui. While Rapanui rights to marine space have yet to be legally established, discussions regarding the possibility for large-scale marine conservation have expanded Rapanui claims to marine space and the right to manage it. This political change process is at least in part a reflection of the scale of the conservation undertaking at stake, and the way Chile's commitments to international agreements on indigenous rights and to marine conservation intersect.

Palau National Marine Sanctuary (Republic of Palau)

Palau passed the Palau National Marine Sanctuary (PNMS) Act in 2015. The PNMS established 80% of Palau's EEZ, approximately 500,000 km^2, as a no-take reserve and designated 20% as a domestic fishing zone (DFZ). Under the law, fish caught in the DFZ "shall only be available for domestic sale" with an exemption allowing export of fish caught through free school purse-seining after landing in Palau (RPPL 9–49 § 164b and c). Fishing in the no-take reserve will be incrementally drawn down and by 2020 the full measures of PNMS will take effect. Here we focus on how the PNMS has initiated interlinked geographic and economic change processes in Palau that are reorganizing the use of its ocean space and its fisheries economy.

Many interviewees view the PNMS as an effort to reclaim Palauan waters and resources for Palauan benefit. Palau's offshore fishery is dominated by foreign tuna vessels. Palau brings in millions (USD) from the sale of fishing rights to foreign vessels fishing both within Palau and, through tradable vessel days allocated to Palau under the Parties to the Nauru Agreement (PNA), outside of Palau in PNA members' EEZs. However, most of the value and the tuna remain outside Palau as fish are sold overseas. Only the lowest-graded tuna are sold in Palau; even here, neither the industrial-scale fishing companies nor vessels are domestically owned, with very few and limited exceptions. Furthermore, Palauan waters are subject to illegal, unreported, and unregulated (IUU) fishing that has been difficult to address under their current capacity. Relative to the foreign fleets, Palau's domestic fishing industry for pelagic fish is small.

President Tommy Remengesau's initial proposal sought to make Palau's entire EEZ a no-take LMPA, excluding the 12-mile territorial sea. A total ban on commercial fishing, however, risked closing off Palau's benefits from the vessel day scheme under the PNA (Aqorau 2015) and other foreign fishing agreements. The President's initial proposal was scaled back in the final PNMS legislation to include a DFZ allowing for free school purse seining and long-lining. The motivations for these changes were multifaceted, but according to interviewees, one motivation was to keep open the possibility that Palau could pursue large-scale marine conservation and continue to sell their allocated vessel days under the PNA for use within the DFZ and/or other PNA country's EEZs. Under the current legislation, however, most interviewees questioned the interest and/or ability of foreign companies/vessels, including both purse seine and the more common long-line vessels, to continue operating in the DFZ given its limited space, the export prohibitions, and the weak Palauan market for high-grade tuna. Many interviewees expressed doubt that foreign vessels will fish in Palauan waters, while some suggested outright that they expect all or nearly all foreign vessels to leave by 2020. With less foreign competition expected, local fishers are being encouraged to expand their offshore operations into the DFZ in order to supply pelagic fish to the growing domestic market and reduce pressure on inshore reefs. To defray potential revenue losses from foreign fishing, the PNMS law also includes a provision to increase visitor fees.

In response to the PNMS legislation, Palau is taking steps to expand its offshore domestic fishing fleet, increase EEZ monitoring and surveillance capabilities, and attract external assistance to do both. Partners including scientists, NGOs, and other nations have already committed in-kind (e.g., boats, technologies, training, expertise) and financial support to bolster Palau's monitoring and surveillance throughout its EEZ to reduce IUU fishing (Terrill et al. 2015). This has already had "a deterrent effect" according to local enforcement officials. The PNMS office is also working on proposals to subsidize development of domestic fisheries and exploring the potential for a trust fund that would buy back Palau's PNA vessel days in an effort to reduce overall regional fishing pressure rather than displace it, while maintaining that revenue stream for Palau. Local fishermen expect that pelagic stocks will expand and also come closer to shore in the DFZ, where they can fish without foreign competition. Conservationists hope that the movement of more local fishermen into the DFZ will reduce pressure on Palau's taxed reefs.

In summary, the PNMS has set in motion interlinked economic and geographic change processes through which Palau is reorganizing and reclaiming its EEZ in an effort to benefit its people. Though our data suggest a range of possible economic and other social impacts that may follow, the future is uncertain. In one scenario, Palau will reap offshore and inshore conservation benefits, grow its domestic fishery, reduce IUU, and maintain some revenue from PNA vessel day sales while increasing revenue from tourist fees. Important questions remain, however, about whether and how a Palauan fleet will expand into the DFZ and what such an expansion would mean for fish prices, fishing pressure, and reef conservation, and how exactly how the PNMS will intersect with the PNA and other fishing agreements. What is clear at this stage is that the PNMS has initiated processes for Palau to undertake national conservation and development simultaneously in an effort to reclaim their ocean space and resources.

Phoenix Islands Protected Area (Republic of Kiribati)

In 2008, the government of Kiribati established the Phoenix Islands Protected Area (PIPA). At 408,250 km^2, PIPA covers more than 10% of Kiribati's EEZ; as of January 1, 2015, PIPA became 99.4% no-take (Rotjan et al. 2014). There are currently only 30 people (government staff and their families) living in the Phoenix Islands group, and the eight atolls within PIPA have not been traditionally inhabited by the I-Kiribati people. Most of the Kiribati population lives in the Gilbert Islands, more than 800 miles west of PIPA. Despite the remoteness of the Phoenix Islands, the establishment of PIPA has led to cultural and political impacts, as I-Kiribati people have interpreted the LMPA in relation to existing cultural values and perceptions of Kiribati as a nation.

First, PIPA has fostered a strong sense of national pride, a political impact (Rotjan et al. 2014). This pride was expressed by many respondents, from government officials and employees to local nonstate actors and the public more broadly. For example, an interviewee living in Tarawa said, "I'm proud of [PIPA]," and described how "a lot of very new people" have come to Kiribati because of the newly established LMPA. In an outreach activity run by the PIPA Implementation Office on an outer island of Kiribati in 2016, PIPA was listed alongside LMPAs in Australia and the United States; upon learning about PIPA during this outreach presentation, a government employee expressed pride in Kiribati for achieving a large conservation feat on par with those of more developed countries. In general, I-Kiribati interviewees were proud of their country for doing something as noteworthy as declaring a LMPA. Some respondents also commented more specifically on their pride in the leadership Kiribati has demonstrated by establishing PIPA, noting that Kiribati was the first country in the Global South to establish an LMPA and that other Pacific Islands have used PIPA as a model for their own conservation efforts. A government employee explained it was "one way of getting us…on the world map, by telling the world that Kiribati is [a] conservation leader by protecting…this big size of ocean." Indeed, as former President Anote Tong explained, "one of the biggest values of declaring the PIPA is… it [has] attracted a lot of international attention."[4] While national pride is itself a political impact of PIPA, many respondents were hopeful that the increased international attention on Kiribati resulting from PIPA would result in additional material benefits, such as revenue from tourism.

A second, cultural impact, has also resulted from PIPA, as some I-Kiribati people have interpreted PIPA in relation to I-Kiribati cultural identity. A wide range of interviewees, including current and former politicians and government employees as well as citizens involved in education and media, reported that PIPA represents a traditional cultural value for ocean conservation, which is important to their identity as I-Kiribati people. As a representative from the Ministry of Culture in Tarawa said, "it is quite good that we protect our natural resources because…if not, then…we will not be identified as I-Kiribati people." One government respondent even suggested that PIPA is promoting a "revival" of this cultural value for conservation by reminding people of "the benefits [of having] a conservation mindset practice." This mindset was described by some interviewees as being vital to maintaining I-Kiribati identity, particularly in a context of increasing population density and demands for economic development. Although their ancestors did not live in the Phoenix Islands, nor do the islands themselves hold particular cultural value, one I-Kiribati respondent explained that because PIPA resembles the environment of their ancestors (with abundant marine life, including culturally important species such as sharks), its protection is culturally symbolic and meaningful.

In summary, as I-Kiribati people learn about PIPA, they interpret the LMPA in relation to their preexisting values and identity. For some individuals, this has resulted in increased national pride or a perception of cultural revival. Both of these impacts may be attributed in part to social change processes initiated in 2012 by the PIPA outreach and awareness program (Rotjan et al. 2014). The outreach program is conducted primarily on the island of Tarawa, and promotes PIPA to the public often through musical performances at local events. Although some interviewees perceived impacts such as national pride and cultural revival as growing in response to outreach efforts, others suggested these impacts have been minimal as public awareness about PIPA is still low, especially outside of Tarawa. However, as Vanclay (2002) argues, social change processes can lead to important perceived impacts that can be felt at any level of social organization; they do not have to be universal. Furthermore, for those who have experienced or perceive cultural and political impacts, they arise despite the distance from and lack of former connection to PIPA.

Marianas Trench Marine National Monument (CNMI and Guam, USA)

Through a presidential proclamation in 2009, U.S. President George W. Bush established the Marianas Trench Marine National Monument (MTMNM) in the CNMI and, to a smaller extent, Guam. The MTMNM comprises 250,487 km^2 of submerged lands and waters including the waters and submerged lands of three remote, uninhabited Mariana Islands (the Islands Unit), and the submerged lands of the Mariana Trench (the Trench Unit) and certain volcanic sites (the Volcanic Unit). Here we focus on how the MTMNM intersected with a longstanding political and legal process regarding sovereignty of submerged lands surrounding the CNMI.

In 1975, the United States and the CNMI signed a covenant establishing the commonwealth status of the CNMI. Shortly thereafter, questions arose about sovereignty over submerged lands and marine resources surrounding the CNMI. The CNMI enacted two laws in 1979 and 1980 to assert their "sovereignty, ownership, and exclusive right to control submerged lands and marine resources underlying a 12-mile territorial sea, as well as a 200-mile exclusive economic zone" (CNMI v. U.S. 2003). A lengthy legal battle ensued, with the U.S. government asserting their "ownership and paramount right to control and regulate the submerged lands and marine resources" (CNMI vs. U.S. 2003). It is difficult to overstate the perceived cultural, political, and economic importance of this territory for the CNMI, which has experienced colonial control since 1564.

In 2007, when Pew began advocating for an LMPA in the CNMI, the U.S. had yet to recognize the CNMI's territorial claims. Key officials in the CNMI government were among the most vocal opponents of the proposed LMPA, contesting what they perceived as a top-down, unilateral designation process under the Antiquities Act of 1906. Despite their opposition, these officials saw that establishing a "blue legacy" for outgoing President Bush was a White House priority, and believed the monument would be designated with or without their support. At the same time, interviewees perceived that the Bush administration sought public approval from the CNMI government even if they didn't legally require it. As one interviewee put it, "Bush didn't wanna appear to be a bully," while a second said "They didn't want this to be controversial." Within this context, the CNMI governor and his advisors saw an opportunity to leverage the LMPA proposal toward their own political ends. As a third interviewee reflected, "Bush was gonna make it anyway, whether we want it or not. So what choice do we have? The choice is [whether to] get some of our ideals." According to another interviewee, once they "saw the writing on the wall," the CNMI gave the White House a list of conditions required for their consent to the monument. One of these conditions was sovereignty over submerged lands throughout the commonwealth. Multiple interviewees involved in the negotiations said that the White House "promised" to meet these conditions.

Six months following the proclamation, the CNMI delegate to the U.S. House of Representatives introduced a bill to convey submerged lands to the CNMI (H.R. 934). A later iteration of the bill was passed in 2013 (P.L. 113–34), providing the CNMI with authority over "the seabed, subsoil, water column and surface water resources in the three-mile coastal zone, including jurisdiction over all mineral, energy and fishing development" (DOI 2015). One interviewee, a senior U.S. federal official, attributed this action to the promises made to the CNMI during the MTMNM negotiations: "as [a] result of the commitments there was a concerted effort, and with positive results toward, submerged lands being granted to the CNMI." A CNMI official we interviewed also recognized this as a possibility. While we cannot establish a causal link, one thing is clear: the CNMI used the MTMNM proposal to leverage high-level U.S. political support for the CNMI's territorial claims.

Tempering this legal victory for the CNMI, however, were limitations the U.S. government would later place on the conveyance of submerged lands within the MTMNM, which limited the CNMI's sovereignty within those spaces and reignited CNMI-U.S. tensions on the issue. Specifically, U.S. President Barack Obama withheld the conveyance of submerged lands within the MTNMN islands unit (a relatively small subset of the total conveyance) until the U.S. federal agencies and the CNMI government agreed upon provisions for their management and protection (Proclamation No. 9077). This was a source of resentment and concern in the CNMI: "These denials of ownership of submerged lands and the creation of the national monument are critical issues affecting the relationship of the CNMI and the U.S." (Governor Ralph D. L. G. Torres in Villahermosa 2016). In September 2016, the CNMI and U.S. federal agencies signed an agreement committing the CNMI government to coordinate management of the submerged lands within the Islands Unit of the MTNMN in accordance with President Bush's initial proclamation. Two months later, a patent transferred the withheld submerged lands to the CNMI. The patent, however, included an easement for the U.S. ensuring that these submerged lands would be protected in perpetuity (Patent Number 2016–1). Although our data were collected in 2015, CNMI-based interviewees who knew about the proposed easement at the time felt slighted and angered by the mistrust that this easement signaled on the part of the U.S. federal government.

In summary, we argue that because the MTMNM was negotiated at the highest political levels, it became enrolled in a longstanding conflict over state sovereignty with mixed outcomes for the CNMI. On the one hand, the MTMNM gave the CNMI leverage (a political impact) within a broader political and legal change process that ultimately culminated in U.S. recognition for the CNMI's territorial claims. On the other hand, the MTNMN would later have legal impacts for the U.S. conveyance, effectively limiting the CNMI's sovereignty over submerged lands within the monument by holding them to a set of management prescriptions set by the U.S. federal government.

The social outcomes of LMPAs: An alternative narrative

This paper engages the SIA literature to offer a conceptual framework for understanding the social outcomes of LMPAs in terms of both social impacts and social change processes. We find the distinction between social impacts and social change processes useful in providing a framework for perceiving a broad range of social outcomes, and in directing attention to processes and perceptions as important outcomes themselves. Social outcomes can be difficult to parse, and in some cases, clear social impacts are still only anticipated or speculated. In these cases, attention to social change processes can signal clear shifts in important social conditions that will affect LMPA trajectories and future outcomes.

We demonstrate the utility of this framework by using it to characterize diverse types of social outcomes in five proposed and designated LMPAs, including: community and institutional impacts from a stalled attempt to designate an LMPA Bermuda; a political change process linked to an LMPA proposal under consideration in Rapa Nui; geographic and economic change processes triggered by a recently designated LMPA in Palau; political and cultural impacts tied to an established LMPA in Kiribati; and political and legal impacts from an established LMPA in the CNMI/Guam. Researchers of conventional MPAs have established the need to consider social outcomes (including perceptions) prior to and during MPA establishment (e.g., Gonzalez and Jentoft 2011), and our results confirm the same is true for LMPAs despite assumptions and practices to the contrary. In Bermuda, a proposed LMPA had an institutional impact on a separate, high-profile international collaboration by greatly reducing the area of collaboration in the Hamilton Declaration regarding conservation in the Sargasso

Sea. In Rapa Nui, an LMPA under consideration advanced a political change process that is expanding indigenous claims to territory through the inclusion of marine space within their negotiations with Chile. Palau is using an LMPA to reclaim its EEZ through national-scale economic and geographical change processes that are directed toward crowding out foreign tuna fishing and developing a national offshore fleet, among other things. Despite its location in a remote place most I-Kiribati people will never see, PIPA has led to perceptions of strengthened national pride and reinforcement of certain aspects of I-Kiribati culture. The MTMNM leveraged political support for the CNMI's longstanding claims to submerged lands, but also ultimately limited their sovereignty gains in those submerged lands included within the monument. By highlighting social outcomes in LMPAs at varying stages, we demonstrate that even remote LMPAs have important social outcomes, and that they emerge as soon as a discussion begins about whether to designate one.

While LMPAs may be similar to conventional MPAs in that both are social spaces that produce social outcomes, our examples suggest that the nature of those outcomes may differ in some respects. The social outcomes of conventional MPAs are typically experienced at a fairly small scale by resource users and communities who are physically close to the space and using resources directly. Our results highlight the potential for LMPAs to impart outcomes at a higher level of social organization, which changes the scope and type of affected populations and, in some cases, the nature and stakes of the outcomes themselves. In Palau, the reorganization of fishing economies and geographies through the PNMS is affecting national and international economies, stakeholders, and spaces. While nearshore MPAs often coincide with incentives to develop offshore fisheries, we are not aware of any examples where a conventional MPA has been leveraged to develop a national fishing fleet, as is the PNMS. The social change processes associated with the PNMS may ultimately affect the entire population of Palau. The institutional impact of removing Bermuda's EEZ from the Geographical Area of Collaboration in the Hamilton Declaration is experienced at an international scale, affecting people in Bermuda but also regional and global actors, and relations among them. The LMPA processes in Rapa Nui and the CNMI have given indigenous and local leaders political leverage and windows through which to advance territorial claims at unprecedented, jurisdiction-wide scales, with potential implications for local and indigenous populations as a whole. While it has been recognized that LMPAs may protect existing "cultural seascapes and long practiced oceanic traditions" (Wilhelm et al. 2014, 24), PIPA demonstrates how a remote LMPA may also affect political and cultural relationships for people who do not have pre-existing material or cultural connections to the particular ocean space being protected. Further, while many conventional MPAs engender pride at a community level (Pollnac et al 2001), few intersect with ideas of the nation and engender feelings of national (as opposed to community) pride as does PIPA. Relative to community pride, national pride may ultimately translate into a host of distinct secondary impacts through, for example, increased public support, legitimacy, and capacity for a national government. Additional work is needed to track the full range of indirect, secondary, and "downstream" social outcomes linked to those we describe here. We suspect they will be farther reaching and qualitatively different from those experienced in conventional MPAs.

We attribute the distinctiveness of the examples examined here to two related characteristics of the LMPA efforts in our study: their geography and their intersection with high-level politics and policy processes. In Rapa Nui, the extensive size of the proposed LMPA gave indigenous Rapanui a position and opportunity to seek rights at an EEZ scale because of its importance to Chile's president, who has made high-profile commitments to both the LMPA and indigenous rights. The situation is similar in the CNMI, where the Bush administration's interest in the MTMNM gave the CNMI high-level political leverage toward longstanding territorial claims. The institutional impact of the Bermuda Blue Halo Initiative resulted from the way in which the proposed LMPA intersected with an international policy process, which was at least in part a reflection of its extensive size and geography within the Sargasso Sea. In Palau, an EEZ-wide LMPA (inclusive of the DFZ) meant that it necessarily figured into national and international conservation and development processes. And finally, interviewees in Kiribati explicitly linked the large size of PIPA to perceptions of national pride in Kiribati as an international conservation leader. In all these cases, the large scale of the proposed or designated MPAs intersected with political priorities and policy processes at national and/or international levels to generate outcomes not typical in smaller MPAs closer to shore.

In summary, this paper concludes that: (1) social outcomes arise even in very remote LMPAs; (2) LMPA efforts generate social outcomes at all stages of development; (3) LMPAs have the potential to produce distinctive social outcomes; (4) the potential for LMPAs to impart distinctive outcomes is a result of their unique geographies and/or intersection with high-level (national and international) politics and policy processes; and (5) social outcomes of LMPAs may be usefully conceptualized in terms of both social change processes and social impacts. It is important that researchers, practitioners, politicians, and donors are attuned to these dynamics as they have the potential to affect the trajectories of LMPAs and far-ranging social systems. Many LMPAs have been strategically designated in so-called "residual" areas in an effort to minimize costs to resource users, and therefore, political costs to the designators (Devillers et al. 2014). There is an assumption that remote spaces with few direct uses present easy political wins (Steinberg 2008). As our results demonstrate, however, resource users are not the only stakeholders to affect and be affected by LMPA negotiations and outcomes. Rather, the geographical and political features of LMPAs gives them the potential to intersect with broader and more diverse populations including but not limited to people with direct, material experiences or uses of the protected spaces. This is not to say that LMPAs will not affect local populations (in fact, all the outcomes we examine have implications for local residents and/or indigenous people), but rather that they may challenge traditional understandings of what it means to be a "local" stakeholder. Key local stakeholders of the LMPAs in Kiribati and Palau, for example, may be simultaneously understood as indigenous and national, and few have a physical or material connection to the space in question. Our findings signal the need for broader conceptualizations of social outcomes of different kinds, at different scales, and/or for different populations in addition to those most familiar based on work in conventional MPAs. As the LMPA movement continues to develop, we hope to see much more attention in practice and research to the full range, magnitude, and distribution of their social outcomes over time.

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Notes

1. At the time of writing, an LMPA in Rapa Nui was under consideration, so we refer to it as "proposed" throughout this paper. Readers should be aware, however, that after the paper was accepted, a multiple use marine protected area was approved on September 4th, 2017 through an indigenous consultation.
2. Except L. Fairbanks, who joined the team later and was trained independently in the shared framework.
3. We thank Jesse Cleary, Marine Geospatial, and Ecology Lab at Duke University, for providing this estimate.
4. President Tong gave us permission to attribute his interview.

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